[Problem 1]

(a) suppose each $y^i = [y_x^i, y_y^i]$, in which y_x^i and y_y^i are known constants

we assume that the rectangle is aligned with x- and y-axis,

and the corner point which is closest to 0 is $A(a_x, a_y)$, and suppose length is l, width is w

$$\begin{aligned} \underset{a_x, a_y, l, w}{\text{minimum}} \quad & l \cdot w \\ \text{subject to} \quad & a_x \le \min(y_x^1, y_x^2, \dots, y_x^K) \\ & a_x + l \ge \max(y_x^1, y_x^2, \dots, y_x^K) \\ & a_y \le \min(y_y^1, y_y^2, \dots, y_y^K) \\ & a_y + w \ge \max(y_y^1, y_y^2, \dots, y_y^K) \end{aligned}$$

and it is a non-linear program

(b) ① It changes the objective function into:

$$\underset{a_x, a_y, l, w}{\text{minimum}} \quad 2(l+w)$$

and change it into a linear program

② The solution is a rectangle with $l = \max(y_x^1, \dots, y_x^K) - \min(y_x^1, \dots, y_x^K)$

$w = \max(y_y^1, \dots, y_y^K) - \min(y_y^1, \dots, y_y^K)$

③ It is the same solution with (a)

suppose there exist a better solution (l^*, w^*) such that $l^* w^* < lw$

then $\exists\ l^* < \max(y_x^1, \dots, y_x^K) - \min(y_x^1, \dots, y_x^K)$

or $w^* < \max(y_y^1, \dots, y_y^K) - \min(y_y^1, \dots, y_y^K)$

then there must be a point out of the rectangle,

so (b)'s solution is the same with (a)

[Problem 2]

(a) suppose for one day, we produce x_1 type I, x_2 type II, x_3 type III

$$\begin{aligned} \underset{x_1, x_2, x_3}{\text{maximize}} \quad & 57x_1 + 54x_2 + 80x_3 \\ \text{subject to} \quad & \tfrac{1}{5}x_1 + \tfrac{1}{4}x_2 + \tfrac{1}{2}x_3 \le 150 \\ & \tfrac{1}{5}x_1 + \tfrac{1}{4}x_2 + \tfrac{1}{4}x_3 \le 80 \\ & x_1 \ge 0 \\ & x_2 \ge 0 \\ & x_3 \ge 0 \end{aligned}$$

(b) $$\begin{aligned} \underset{x_1, x_2, x_3}{\text{maximize}} \quad & 57x_1 + 54x_2 + 80x_3 - \left(\tfrac{x_1}{5} + \tfrac{x_2}{4} + \tfrac{1}{2}x_3 - 150\right) \times 50 \\ \text{subject to} \quad & \tfrac{1}{5}x_1 + \tfrac{1}{4}x_2 + \tfrac{1}{2}x_3 \le 210 \\ & \tfrac{1}{5}x_1 + \tfrac{1}{4}x_2 + \tfrac{1}{4}x_3 \le 80 \\ & x_1 \ge 0 \\ & x_2 \ge 0 \\ & x_3 \ge 0 \end{aligned}$$

(c) Python code：

```
import cvxpy as cp

x1 = cp.Variable(integer=True)
x2 = cp.Variable(integer=True)
x3 = cp.Variable(integer=True)

objective = cp.Maximize(57*x1 + 54*x2 + 80*x3)

constraints = [
    (1/3)*x1 + (1/4)*x2 + (1/2)*x3 <= 150,
    (1/5)*x1 + (1/4)*x2 + (1/4)*x3 <= 80,
    x1 >= 0,
    x2 >= 0,
    x3 >= 0
]
problem = cp.Problem(objective, constraints)

problem.solve()
```

Optimal solution: x1 = 0, x2 = 40, x3 = 280
Optimal value: 24560

[Problem 3]

We let the cost of moving iron rod between each pair be W:

$$W=\begin{bmatrix} w_{11} & w_{12} & \cdots & w_{15} \\ w_{21} & w_{22} & & w_{25} \\ \vdots & & \ddots & \vdots \\ w_{51} & w_{52} & \cdots & w_{55} \end{bmatrix}$$

and for the iron rod transferied from i to j, we denote it as x_{ij}, and the initial value of each company is x_i

$$X=\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{15} \\ x_{21} & x_{22} & & x_{25} \\ \vdots & & \ddots & \vdots \\ x_{51} & x_{52} & \cdots & x_{55} \end{bmatrix}$$

$$\begin{aligned} \underset{x_{ij}}{\text{minimum}} \quad & \sum_{i=1}^{5}\sum_{j=1}^{5} w_{ij}\cdot x_{ij}\cdot 0.1 \\ \text{subject to} \quad & 7000-\sum_{j=1}^{5}x_{1j}+\sum_{j=1}^{5}x_{j1}\geq 0 \\ & 6000-\sum_{j=1}^{5}x_{2j}+\sum_{j=1}^{5}x_{j2}\geq 2000 \\ & 4000-\sum_{j=1}^{5}x_{3j}+\sum_{j=1}^{5}x_{j3}\geq 500 \\ & -\sum_{j=1}^{5}x_{4j}+\sum_{j=1}^{5}x_{j4}\geq 8000 \\ & 500-\sum_{j=1}^{5}x_{5j}+\sum_{j=1}^{5}x_{j5}\geq 6000 \\ & x_{ij}\geq 0 \quad i,j\in\{1,2,3,4,5\} \\ & x_{ij}\in\mathbb{Z} \quad i,j\in\{1,2,3,4,5\} \end{aligned}$$

```
cost_matrix = [0, 5, 3, 9, 6;
5, 0, 9, 7, 4;
3, 9, 0, 4, 6;
9, 7, 4, 0, 8;
6, 4, 6, 8, 0];

cvx_begin
variables x(5,5)
minimize sum(sum(cost_matrix.*x)* 0.1)
subject to

7000+x(2,1)+x(3,1)+x(4,1)+x(5,1)-x(1,2)-x(1,3)-x(1,4)-x(1,5) >= 0;
6000+x(1,2)+x(3,2)+x(4,2)+x(5,2)-x(2,1)-x(2,3)-x(2,4)-x(2,5) >= 2000;
4000+x(1,3)+x(2,3)+x(4,3)+x(5,3)-x(3,1)-x(3,2)-x(3,4)-x(3,5) >= 500;
x(1,4)+x(2,4)+x(3,4)+x(5,4)-x(4,1)-x(4,2)-x(4,3)-x(4,5) >= 8000;
500+x(1,5)+x(2,5)+x(3,5)+x(4,5)-x(5,1)-x(5,2)-x(5,3)-x(5,4) >= 6000;
x>= zeros(5,5);

cvx_end
```

The **optimal solutions** are x13 = 4500 ; x15 = 1500 ; x25 = 4000 ; x34 = 8000, with the rest xij = 0.

The **optimal objective value** is 7050

[problem 4]

(a) Since the two dogs can't be seperated, for the 12 items, we suppose

$W = [0.3, 0.5, 0.6, 1, 1, 1.2, 1.2, 1.2, 2, 3.5, 6, 8]$

Let x_i be the ith item belong to Bela, y_i be it belong to Daniela, z_i be it belong to Cassandra

$$\underset{x_i, y_i, z_i}{\text{minimize}} \quad \sum_{i=1}^{12} w_i x_i - \sum_{i=1}^{12} w_i z_i$$

$$\text{subject to} \quad \sum_{i=1}^{12} w_i x_i - \sum_{i=1}^{12} w_i y_i \geq 0$$

$$\sum_{i=1}^{12} w_i y_i - \sum_{i=1}^{12} w_i z_i \geq 0$$

$$x_i + y_i + z_i = 1 \quad \forall i$$

$$x_i \geq 0 \quad \forall i$$

$$y_i \geq 0 \quad \forall i$$

$$z_i \geq 0 \quad \forall i$$

$$x_i, y_i, z_i \in \mathbb{R}$$

(b) No. Optimal solution is Cassanda has 8.9, Daniela has 8.8, Bela has 8.8

For Cassanda = 3.5 + 2 + 1.2 + 1.2 + 1 = 8.9

Daniela = 8 + 0.3 + 0.5 = 8.8

Bela = 6 + 1.2 + 1 + 0.6 = 8.8

since it is impossible to equally distribute,

the difference of 0.1 is the optimal value.

[Problem 5]

Suppose we have each room $V = \{A, B, C, D, E, F, G, H, I, J\}$ E represent the doors through rooms

$$\text{minimize} \sum_{v \in V} x_v$$

$$\text{subject to} \quad x_u + x_v \geq 1 \quad \forall (u, v) \in E$$

$$x_v \in \{0, 1\} \quad \forall v \in V$$

since we want to minimize the guards at the doors, we just need to garentee that there be at least one guard in every two rooms, and if the guard is at the door, then it can be count as 1. otherwise the position of the door is counted as 0.